

Google Classroom Code Section A: **22ngzxcq**
Google Classroom Code Section B: **2elb2tll**
Google Classroom Code Section C: **yaksdbfh**

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Course Objective:

This lab course complements the Operating Systems course by providing hands-on practice with its core concepts. Students will explore the power of the Linux terminal and command-line interface, implement system calls, and engage in activities like process creation, management, and interprocess communication. The exercises are designed to strengthen their understanding of how modern operating systems implement various algorithms, with coding activities carried out primarily in C++. Students will also apply their existing knowledge of data structures and related concepts to see their practical use in operating system algorithms.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- **PLO4: Design/Development of Solutions**
- **PLO5: Modern Tool Usage**
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

CLO–PLO Mapping with BT Levels

CLO	Learning Outcome	BT Level	Mapped PLOs
CL01	Practice Linux commands and utilities to gain proficiency with the command-line interface and use it effectively for system management tasks, basic scripting, software installation, and other essential operations in a Linux environment.	A5	PLO5
CL02	Implement system calls, create processes, solve synchronization problems by applying synchronization mechanisms like semaphores and mutexes.	C3	PLO4
CL03	Apply core operating systems algorithms for CPU Scheduling, Disk Scheduling, Memory Management and Deadlock Avoidance.	C3	PLO4

Recommended Books: Soft Copies available

A. Operating System Concepts by **Abraham Silberschatz**, Peter B. Galvin, and Greg Gagne, Wiley; 10th edition

B. Operating Systems: Internals and Design Principles by William Stallings, Prentice Hall; 9th Edition

C. Operating Systems: Three Easy Pieces

Course Outline and Schedule (Tentative) :

Week No.	Topic	Lab No.	Contents	Relation with CLO	Assessment's
W1. 22-Sep	Introduction to the Linux Operating System and CLI	L1.	- Introduction to Linux OS and its distributions. - Different ways of setting up the Linux Environment on the machine. - Terminal, Shell, CMD - Navigating through the file system using linux commands. - VIM Editor in Linux.	CL01	
W2. 29-Sep	Linux Commands and Utilities: Beyond Basics	L2.	- Package Managers for installing and updating softwares using CLI. - Executing C++ Programs from the terminal. - Bash Scripting - File Permission System in Linux. - Remote SSH for establishing secure connections between clients and servers. - Understanding UNIX Philosophy.	CL01	
W3. 06-Oct	System Calls in Operating Systems	L3.	- Implementation of Fork() system call - Implementation of Exec() system call - Implementation of Wait() system call - Implementation of Exit() system call	CL02	
W4.	Process creation and management	L4.	- Leveraging system calls for creating and managing	CL02	

13-Oct			processes using C++ - - Implementation of Threading and Multi Threading using C++		
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W5. 20-Oct	Inter-process communication	L5.	- Inter-process communication using pipes - Inter-process communication using message queues - Inter-process communication using shared memory. - Sockets - RPCs	CL02	
W6. 27-Oct	Process Synchronization	L6.	- Using semaphores to solve process synchronization problems. - Using semaphores to avoid race conditions. - Using semaphores to handle critical sections .	CL02	
W7. 03-Nov	Producer Consumer problem	L7.	- Implementing solution for famous producer consumer problem in operating systems using C++	CL03	

W8. 10-Nov	Reader Writer Problem	L8.	- Implementation of Reader-writer problem in operating systems using C++.	CLO3	
W9. 17-21 Nov	Mid-Term	L9	- Exam	CLO1,CLO2,CLO3	Mid-Term 1
W10. 24-Nov	CPU Scheduling algorithms	L10	- Implementation of FCFS in a simulated fashion using C++ - Implementation of SJF in a simulated fashion using C++ - Implementation of Round Robin in a simulated fashion using C++ - Implementation of Priority Scheduling algorithm in a simulated fashion using C++	CLO3	
W11. 01-Dec	Banker's Algorithm	L11	- Implementation of Banker's Algorithm for avoiding deadlock detection.	CLO3	
W12. 08-Dec	Memory allocation strategies	L12	- Implementation of Best Fit algorithm for memory allocation. - Implementation of First Fit algorithm for memory allocation. - Implementation of Worst Fit algorithm for memory allocation.	CLO3	

W13. 15-Dec	Page replacement algorithms	L13	- Implementation of LRU page replacement algorithm. - Implementation of FIFO page replacement algorithm. - Implementation of Optimal page replacement algorithm.	CL03	
W14. 22-Dec	Disk scheduling algorithms (FCFS, SSTF, SCAN, C-SCAN)	L14	- Implementation of FCFS disk scheduling algorithm. - Implementation of SSTF disk scheduling algorithm. - Implementation of Scan disk scheduling algorithm. - Implementation of C-SCAN disk scheduling algorithm.	CL03	
W15. 29-Dec	Case study: Linux file system vs Windows file system	L15	- Understanding the differences between Windows and Linux file systems. - Appreciating the security aspects of the Linux file system. - File system implementation basics.	CL01	
W16. 05-Jan	Project Presentations	L16	- Project Presentation	CLO1, CLO2, CLO3	
W17. 10-13 Jan	Quiet Week		Quiet Week		Quiet Week
W18. 14-24 Jan	Theory Final exam		- Theory Final exam		
29 Jan	Exam Scripts Shown		-		
30 Jan	Declaration of Result		-		

Assessment Criteria (100%)

Assessment Description	Weight (%)	CLOs	PLOs
Quizzes	10%	CLO1, CLO2, CLO3	PLO4, PLO5
Assignments	15%	CLO1, CLO2, CLO3	PLO4, PLO5
Labs	25%	CLO1, CLO2, CLO3	PLO4, PLO5
Mid Term Exams	25%	CLO1, CLO2, CLO3	PLO4, PLO5
Project	25%	CLO1, CLO2, CLO3	PLO4, PLO5

GRADING TABLE ABSOLUTE SCALE if no. of students in Final Exam are up to 20, otherwise RELATIVE SCALE			
Range	Percentage	Points	Grade
85.00-100	85.00	4.00	A
81.50-84.99	81.50	3.67	A–
78.00-81.49	78.00	3.33	B+
74.50-77.99	74.50	3.00	B
71.00-74.49	71.00	2.67	B–
67.50-70.99	67.50	2.33	C+
64.00-67.49	64.00	2.00	C
60.50-63.99	60.50	1.67	C–
57.00-60.49	57.00	1.33	D+
50.00-56.99	50.00	1.00	D
0.00-49.99	Below 50	0.00	F